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Effects of using a second-screen application on attention, learning, and user experience in an educational content

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Effects of using a second-screen application on attention, learning, and user experience in an educational content

Using a secondary device while viewing a primary device (i.e., TV), or media multitasking, is now common. Numerous researchers and practitioners have attempted to introduce secondary devices into education as a new learning environment providing additional information to the user. However, the learning-related effects of using a second screen remain controversial. This study examined the effects of second-screen-application use on attention, learning performance, and user experience per content relevance for three primary contents (i.e., videos) requiring different amounts of cognitive load: low, medium, and high. Second-screen use reduced learning performance and user satisfaction when the primary content required high cognitive load. However, participants exhibited increased learning performance, concentration, and satisfaction with the primary content requiring medium-cognitive-load when highly relevant information was presented on the secondary screen simultaneously. Based on these findings, guidelines were suggested for designing a second-screen application without degrading users' learning and experience.

Keywords: second-screen application; media multitasking; educational content; content relevance; multimedia learning

Introduction

A market research company (Nielsen Company, 2013) recently reported that 76% of tablet owners searched for general information, 68% surfed the Internet, and 53% used social networking services (SNSs) while watching TV. Further, users' visual attention switches four times per minute between TVs and laptops (Brasel & Gips, 2011) and 27 times per hour between TVs and mobile devices (Steinberg, 2012). This behavior, also known as media multitasking (Carrier, Cheever, Rosen, Benitez, & Chang, 2009), is prevalent in everyday life, particularly in the younger generation (Rideout, Foehr, & Roberts, 2010).

Numerous cognitive scientists and psychologists identified some adverse effects of media multitasking on learners in educational settings. Media multitasking and attention switching are known to negatively influence TV recognition (Lee, Lin, & Robertson, 2011; Oviedo, Tornquist, Cameron, & Chiappe, 2015), recall and comprehension (Armstrong & Chung, 2000; Laine-Hernandez et al., 2014; Srivastava, 2013; Van Cauwenberge, Schaap, & Roy, 2014), reading performance (Lin, Robertson, & Lee, 2009), homework performance (Zhang, Jeong, & Fishbein, 2010), text message comprehension (Jeong & Hwang, 2012), video quality perception (De Meulenaere, Staelens, Broeck, Turck, & Demeester, 2014), brand recognition and recall (Jensen, Walsh, Cobbs, & Turner, 2015; Voorveld, 2011), and memory of lecture content (Hembrooke & Gay, 2003).

Meanwhile, the media and advertisement industries have attempted to develop means of engaging multi-device users in a single service platform, supporting media multitasking behavior. Second-screen applications offer such a platform, providing "a companion experience" (The 2nd Screen Society, 2012) in which users engage in supplementary and relevant content on a secondary device (e.g., smartphone or tablet) while accessing the main content on a primary device (e.g., TV or computer). There are two differences between conventional and second-screen media multitasking applications. First, secondary content (SC) via second-screen applications is temporally synchronized with primary content (PC). A certain piece of SC in a second-screen application is designated for display on the secondary device at a certain time relative to PC whereas SC in conventional media multitasking is completely independent of PC. For example, in second-screen applications, a figure and detailed description of the Earth's internal structure is presented on a tablet while a lecturer discusses Earth on a primary screen. Unlike in second-screen applications, homework completed on a laptop is temporally independent of TV-presented video in conventional media multitasking. Second, in second-screen applications, SC is synchronized contextually with PC. In conventional media multitasking, two irrelevant pieces of content are often consumed (e.g., watching TV while reading). In contrast, second-screen applications essentially assume contextually relevant content (e.g., football player scores on TV; tablet displays detailed player information).

Researchers have attempted to develop various service frameworks (Anstead, Benford, & Houghton, 2014; Basapur et al., 2011; Lee, Yoo, & Han, 2015) and interaction techniques (Cruickshank, Tsekleves, Whitham, Hillm, & Kondo, 2007; Neate, Jones, & Evans, 2015) for second-screen applications. Moreover, several studies have described visual attention distribution between primary and secondary screens (Breidbach, Chandler, & Maglio, 2015; Brown et al., 2014; Holmes, Josephson, & Carney, 2012; Kusumoto, Kinnunen, Kätsyri, Lindroos, & Oittinen, 2014), usage patterns (Ainasoja, Linna, Heikkilä, Lammi, & Oksman, 2014; Courtois & D'heer, 2012), motivation types (Chronister, 2014; Shim, Oh, Song, & Lee, 2015), and watching experiences (Anstead et al., 2014; Basapur et al., 2011; Geerts, Leenheer, De Grooff, Negenman, & Heijstraten, 2014; Kusumoto, Kinnunen, Kätsyri, Lindroos, & Oittinen, 2014; Vinayagamoorthy, Allen, Hammond, & Evans, 2012) in commercialized second-screen applications. However, most of these focused on behavioral characteristics and emotional experience of entertainment content (e.g., sports, reality shows, and dramas) rather than complex cognitive process (e.g., learning).

In educational context, Shin, An, and Kim (2015) reported that the simultaneous use of multiple devices led to higher anxiety and lower competence levels than did sequential use during learning. However, this study did not take into account the influence of content relevance and temporal synchronization between PC and SC. It is noteworthy that very little research exists about the effects of second-screen application on learning, considering temporally and contextually synchronized PC, SC, and their relevance. Unlike conventional media multitasking, in which types of content, tasks, and media are completely regulated by the users' intention, second-screen media multitasking can be designed by content providers and designers producing PC and SC. If certain criteria exist for designing second-screen applications, content providers and designers can determine content-related properties to minimize the negative effects and maximize the positive effects of second-screen application in educational context.

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Therefore, the effects of content-related properties of second-screen application should be investigated to deduce such criteria.

The present study explored the effects of second-screen-application use on learning performance, attention, and user experience. Hypotheses regarding users' behavior in a second-screen application setting were formulated using the basic cognition model of human information processing and evaluated using three educational PC types requiring different cognitive load levels: low, medium, and high. The experiment aimed to validate the effects of content relevance with temporally and contextually synchronized PC and SC in a physically separated dual-screen environment, which is close to an actual second-screen application environment. The implications of and design guidelines for educational second-screen applications and the study's contributions and limitations are presented below.

Theoretical background and research hypotheses

Users' complex learning processes while using second-screen applications can be explained based on information processing and human multitasking theories. First, the limited capacity hypothesis (Baddeley, Scott, Drynan, & Smith, 1969) and model (Lang, 2000, 2006) of human cognitive resources explain how humans process information from early to later cognitive stages. In early cognitive stages, human sensory modules (e.g., eyes and ears) perceive information with visual and auditory attention. In later cognitive stages (e.g., memory and comprehension), as the human brain cannot process all perceived information concurrently, it selects and processes only a subset of information and discards unselected information (Kahneman, 1973; Lang, 2000, 2006). Based on these fundamental theories, Mayer and Moreno (1998) further reported the split-attention effect whereby the display of various types of visual information requiring the same single modality distracts users and inhibits information processing. Meanwhile, studies have shown that human multitasking results in significant effects on humans' cognition and attention. Others have reported that performing dual-task increases interference between visual attention and working memory (Dell'Acqua & Jolicoeur, 2000; Jolicoeur, 1999; Jolicoeur, & Dell'Acqua, 1999). In the context of media multitasking, Jeong and Hwang (2016) reported that media multitasking behavior negatively influences attention and cognitive outcomes (e.g., memory and comprehension), and Wang et al. (2012) showed that the fixation rate and duration of visual attention to the primary task decreases during media multitasking. Therefore, if too much information is provided in multitasking context, people fail to focus on and learn some types of content. This interferes with effective information processing and is critical in second-screen applications, in which learners encounter PC and SC simultaneously on two physically separated screens. Therefore, information overload due to second-screen use more greatly affects learning performance than in single-screen environments. Accordingly, we formulated the following hypothesis:

H1: Second-screen application use will reduce PC learning.

Cognitive overload from excessive information provision occurs more easily when PC (i.e., in a video) includes a large rather than small amount of information.

Kahneman (1973) suggests that information from more than two sources may interrupt effective information processing with introducing the notion of capacity interference and structural interference. Capacity interference occurs when information from more than two sources competes for limited capacity of cognitive resource, and structural interference appears when information from multiple sources requires same sensory channel. Jeong and Hwang (2015) have shown that structural interference has more negative impact on reading comprehension than capacity interference in the context of media multitasking. They reported that learners' reading comprehension on text message is negatively influenced by additional visual information, which results in structural interference, but not by additional auditory information. Unlike the other content requiring single sensory channel, viewing a video (i.e., PC) is a complicated cognitive process requiring the ability to process visual and auditory information simultaneously (Anderson, 1983), and a learner experiences considerable cognitive load when the video is presented with additional visual information, such as slides, in a single screen (Homer, Plass, & Blake, 2008). Such cognitive processes and load vary significantly with video pacing (Lang, Bolls, Potter, & Kawahara, 1999), structural complexity, and information density (Lang, Bolls, Potter, & Kawahara, 1999; Lang, Park, Sanders-Jackson, Wilson, & Wang, 2007; Lang, Kurita, Gao, & Rubenking, 2013). When videos present information with high speed and complex structure, viewers easily experience cognitive overload. Furthermore, Shah and Freedman (2003) reported that amount of visualized information presented (e.g., graphs, maps, and

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tables), which demands interpretation and comprehension ability, is a critical influence factor causing cognitive load. Moreover, Salvucci and Bogunovich (2010) has shown that multitasking is better performed under low than under high cognitive load. Therefore, we inferred that the effects of additional provision of SC on learning could differ per the amount of information already contained in PC, which requires users' cognitive resources. Accordingly, we formulated the following hypothesis:

> **H2**: The negative effects of second-screen-application use on PC learning will increase with the amount of cognitive load that PC requires.

In contrast, some studies have shown that additional information enhances information processing. For example, perception and memory of words presented in a sentence are greater than for the same word presented alone (Tulving, Mandler, & Baumal, 1964). The richer context provided by the sentence facilitates deeper meaning processing and memory retrieval. However, this effect differs per content relevance (e.g., Stroop, 1935). Unrelated or incongruent information requires greater cognitive resources than when no additional information is provided. Furthermore, the coherence principle (Mayer, 2005) in multimedia learning theory suggests that instructional information in a certain modality combined with attractive but irrelevant additional information in the same modality distracts the learner and diminishes learning by increasing cognitive demand to process both types of information simultaneously. In other words, combining two highly relevant types of multimedia information facilitates learning by reducing cognitive load (Park, Moreno, Seufert, & Brünken, 2011).

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Salvucci and Taatgen (2008) proposed the threaded cognition theory, which explains sequential and concurrent multiple processing of each cognitive resource (e.g., visual, manual, etc.) with multiple "threads" in a single timeline to simulate and formulate the human multitasking. The threaded cognition theory has been shown that human multitasking can be effectively performed when multiple tasks share common goals and do not require same cognitive resource at the same time. In the context of media multitasking, Wang, Irwin, Cooper, and Srivastava (2015) presented the notion of task relevance, which refers "whether the tasks involved in media multitasking serve closely related goals", adapting the threaded cognition. In addition, Jeong and Hwang (2016) reported that the task relevance between multiple tasks significantly moderates media multitaskers' cognitive outcomes (e.g., factual recall, comprehension); cognitive outcomes decrease when two tasks are irrelevant. Similarly, in second-screen context, learning could depend on the difference in levels of contextual relevance of SC to PC. Therefore, we inferred that differences in contextual relevance of SC to PC could either reinforce or interfere with PC retrieval. Accordingly, we formulated the following hypothesis:

> **H3**: Second-screen application use when SC relevance is low will exert a greater negative effect on PC learning relative to that exerted when SC relevance is high.

As subjective experience of online learning is crucial in determining usage intention (Roca, Chiu, & Martnez, 2006), it is important to examine emotional factors involved in second-screen-application use. Certain learning- and PC-related emotions

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(e.g., concentration level and satisfaction with PC) and self-confidence regarding learning outcomes could serve as influential factors determining the overall experience in an educational context. Several educational researchers found learners' concentration to be a major indicator determining learning performance (Loranger, 1994; Albaili, 1997; Entwistle, 1998; Yip & Chung, 2005), and learners' perceived satisfaction should be considered in evaluating e-learning environment effectiveness (Zhu, 2012; Liaw & Huang, 2013). Furthermore, among various emotional factors influencing learning, recent studies reported that self-confidence of learners highly predicts actual learning achievement (Stankov, Lee, Luo, & Hogan, 2012; Komarraju & Nadler, 2013; Morony, Kleitman, Lee, & Stankov, 2013; Stankov & Lee, 2014). In the context of media multitasking, several recent studies have reported that media multitasking positively influences multitaskers' attitudes by distracting counterarguments (Jeong & Hwang, 2012, 2015), unlike learning (Jeong & Hwang, 2016). Accordingly, we formulated the following hypothesis:

> **H4**: Second-screen application use will have positive effect on user experience (e.g., concentration, satisfaction, and confidence).

Method

Participants and design

Forty-six participants (mean age = 23.9 years, SD = .38; 8 female) participated in an hour-long experiment and were compensated with $20 each. The number of participants was determined based on a priori power analysis (Cohen's d: 0.8; desired power: 0.8; probability level: .05; one-tailed t-test) and sample size in previous studies about media multitasking (Armstrong & Chung, 2000; Fante, Jacobi, & Sexton, 2013; Gamble, Howard Jr, & Howard, 2014; Hembrooke & Gay, 2003; Sana, Weston, & Cepeda, 2013; Tran & Subrahmanyam, 2013) to investigate the main effects of second-screen usage. The analysis suggested the minimum sample size as 42 (21 for each screen-group), and we recruited 52 participants considering about 10–15% outliers. The results were analyzed based on the data from 46 participants, excluding 6 outliers. Participants were unaware of the study purpose and possessed no background knowledge regarding the experimental materials. Eighteen were undergraduate students, and 28 were graduate students, all with engineering backgrounds. They were randomly assigned to single- (n = 20; 4 female) and dual-screen groups (n = 26; 4 female). The experiment used a 3 (cognitive load) × 3 (content relevance) × 2 (screen group) design. Cognitive load (low, medium, and high) and content relevance (low, medium, and high) were within-subjects factors, and screen group (single and dual screen) was a between-subjects factor.

Apparatus

The experimental environment, which involved a web-based second-screen framework (Lee et al., 2015), consisted of a video streaming server to play PC, a content server to present SC at a designated point during PC presentation, and a synchronization server to synchronize PC and SC (Figure 1[a]). PC was displayed on a 27-inch desktop monitor and SC on a 9.7-inch tablet (Figure 1[b]). An additional webcam tracked participants' visual attention using a face detection algorithm (Bos, 2011).

[Insert Figure 1 here.]

Stimulus materials

We recruited five trained graduate students as assessors to select and assess online educational videos as PC stimuli. PC selection was based on recorded language, genre, theme (e.g., history and liberal arts), level of difficulty, background (e.g., engineering), lack of offensiveness (i.e., no racial or other type of discrimination), and video duration, referencing Berk's (2009) criteria for selecting videos as instructional tools. The assessors also selected videos with typical plot structures and elements (e.g., a lecturer with a blackboard in the background for the lecture video) to maintain generality.

Three videos including different amounts of information were selected to vary cognitive load, referring to Shah and Freedman's (2003) and Lang et al.'s (2007) research. Levels of cognitive load were determined by analyzing the overall amount of information in both visual and auditory domains of each video, because cognitive load is resulted from effortful learning, which is achieved by successful integration of visual and auditory information at working memory of a learner throughout playing timeline not by either visual or auditory information alone. In the high-cognitive-load condition, the lecture video contained large amounts of visual (18 figures) and auditory (1.56 words per second) information. In the low-cognitive-load condition, the documentary video included small amounts of visual (four figures) and auditory (1.11 words per second) information. In the medium-cognitive-load condition, the conference presentation video contained a large amount of auditory (1.73 words per second) information, which was more similar to the high-cognitive-load video than the low-cognitive-load video, and a small amount of visual information (three figures), which was more similar to the low-cognitive-load video than the high-cognitive-load video. The amount of visual information for each cognitive load condition was based on the number of figures (e.g., tables, charts, and maps), which required interpretation and comprehension ability. The amount of auditory information for each cognitive load condition was calculated using the number of words per second in audio narration.

PC stimuli included three online video clips (approximately 15 min each) selected from different content categories (i.e., lecture, documentary, and conference presentation videos) and contained different topics to maintain homogeneity across each participant and to avoid prior knowledge generation as each participant watched all three videos in a single experiment. The lecture video included specific historical facts related to a current political issue regarding Dok-do Island. The documentary included information regarding the rise and fall of the Ottoman Empire during the 16th and 17th centuries. The conference presentation video included a speech about the social and philosophical meanings of happiness. Although these videos were selected from different categories and contained different topics, each contained high school-level information, and professional knowledge was not required to understand the presented terms and entire storyline.

The following nine types of SC were prepared for each piece of PC:

- Personal information: information about a person present in the PC
- Related news: online news about the PC
- Academic reference: summary of academic work related to the PC
- Background information: geographical, demographic, statistical, and historical information about the PC
- Episode list: list of other episodes related to the PC
- Advertisement: commercial advertisement of a product or service related to the PC
- Price information: price information for a product or service related to the PC
- Common SNS reactions: ordinary people's reactions to SNSs concerning the PC
- Celebrities' SNS reactions: opinion leaders' SNS reactions concerning the PC

SC type was determined based on the current trend in massive open online courses, which focused on embedding not only complementary learning material but also the social networking system and commercial advertisements on a single-service platform (Taneja & Goel, 2014). Each SC type was designed for display at contextually and temporally appropriate points that would not make participants feel awkward during presentation of corresponding PC. For example, when a presenter in the conference presentation video explained the content of a book, purchasing information regarding the book was displayed on a tablet.

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SC relevance to PC was preliminarily evaluated by 65 respondents via an online questionnaire. The questionnaire targeted respondents' opinions regarding contextual relevance of each piece of SC to each piece of PC without information regarding the second-screen application (e.g., "Do you think the episode list for the documentary program is contextually related to the documentary video?"). Respondents indicated their opinions using a seven-point Likert scale ranging from 1 (irrelevant) to 7 (highly relevant). SC was categorized into three levels per content relevance (i.e., high, medium, and low) (Table 1).

[Insert Table 1 here.]

Procedure

The experiment was conducted in an isolated room to prevent interference. Before the experiment, participants were informed of the experimental procedure. Each participant completed a basic preliminary questionnaire regarding demographic information and media multitasking patterns and preferences and was randomly assigned to either a single-screen or dual-screen group. Participants completed the experiment sitting in front of a desktop computer using a headset.

Figure 2(a) shows the detailed experimental procedure for the single-screen group. It consisted of three sessions, each corresponding to a cognitive load level, ordered randomly for each participant, to counterbalance experimental condition and avoid recency and primacy effects (Murdock, 1962). In each session, participants watched a video on the desktop monitor and completed a test targeting their learning

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and experience of the video. Participants answered questions concerning learning performance within limited but adequate time (i.e., 1 min per question). The questions targeting user experience were answered without time constraint.

The dual-screen group followed a similar procedure with three differences (Figure 2[b]). First, unlike the single-screen group, dual-screen participants used an additional screen device (i.e., a tablet) on the desk. In each session, participants watched and retrieved nine pieces of SC on the tablet while viewing a video on the desktop monitor. They were also instructed to switch their visual attention freely between screens. Second, the dual-screen group performed an additional dummy test concerning SC, with additional time, to prevent sole concentration of their visual attention on the desktop monitor. The test questions concerning SC were excluded from further analysis. Third, an auditory cue was embedded for each SC video to alert participants to new SC on the tablet as auditory cues are most effective for informing users of SC transition (Neate et al., 2015).

[Insert Figure 2 here.]

Measures

Demographic information and preference survey

The preliminary questionnaire focused on variables pertaining to demographic information, preference, and learning ability to verify participant randomization as control variables. Media multitasking frequency (M = 4.2, SD = 1.327) was measured using a seven-point Likert scale ranging from 1 (never) to 7 (always), via which participants indicated their frequency of consuming multiple content simultaneously, as suggested by Collins (2008), Srivastava (2013), and Van Cauwenberge et al. (2014). Technology acceptance (M = 4.65, SD = 0.82) was measured using a seven-point Likert scale ranging from 1 (unacceptable) to 7 (highly acceptable), which identified participants' preferences for new information devices and technology. Self-rated concentration (M = 4.07, SD = 0.97), comprehension (M = 4.91, SD = 0.84), and factual recall (M = 4.52, SD = 0.75) ability were measured using a seven-point Likert scale ranging from 1 (very poor) to 7 (very good), per Maki and Maki's (2002) procedure.

Visual attention

We used a simple face detection algorithm (Bos, 2011) with a generic webcam on the desktop monitor to measure participants' visual attention to each screen during viewing. The binary event was logged at 10 Hz to determine whether participants were watching the vertically placed desktop monitor or horizontally placed tablet. Each event was synchronized with the PC timeline. Logged event data were processed using a low-pass filter and averaged for all dual-screen participants per the PC timeline. Previous studies (Brown et al., 2014; Holmes et al., 2012; Kusumoto et al., 2014) used eye trackers to monitor visual attention in conventional media multitasking and second-screen applications. The face detection method caused less viewing discomfort and physical fatigue than observed with desk- or head-mounted eye trackers but guaranteed sufficient precision for the study purpose; in the second-screen context, visual attention toward the primary or secondary screen was detected via face detection.

Learning performance

Participants' learning performance was measured using multiple-choice, short-answer, short descriptive, and sequence-of-events questions. Questions concerning PC (12, 13, and 14 questions for lecture, documentary, and conference videos, respectively) were created based on multimedia learning theory (Mayer, 2009; Mayer & Moreno, 1998; Mayer & Moreno, 2003) to measure memory or comprehension. Memory was measured using multiple choice and short-answer questions pertaining to specific information provided at a certain point during PC presentation (e.g., "What was the name of the Sultan of the Ottoman Empire who conquered Rhodes Island, defeating the Knights of Malta in 1521?"). The short descriptive and sequence-of-events questions targeting comprehension required inference or logical reasoning based on information provided at a certain point during PC presentation (e.g., "The presenter says that the human mind is totally dependent on the 'environment.' Why is 'environment' the most critical factor influencing happiness? Please answer based on the presenter's speech."). As each piece of SC was scheduled for presentation relative to the associated PC timeline, each question was scored in terms of every independent variable (i.e., levels of cognitive load and content relevance) by identifying the relevant moments of the required information. For example, a question in a test for a low-cognitive-load video asked about "the name of the Sultan of the Ottoman Empire who conquered Rhodes Island, defeating the

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Knights of Malta in 1521." A correct answer was presented for a certain duration of the low-cognitive-load video. During the duration, single-screen participants viewed only the low-cognitive-load video, and dual-screen participants viewed the low-cognitive-load video and SC with medium relevance simultaneously. Therefore, the correctness of the question indicates learning performance, and the effect of second-screen use on learning performance can be measured by comparing the learning performance of both groups. All items were scored as either "0" (incorrect) or "1" (correct).

User experience

Participants' experience was measured via five questionnaire items answered on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). These questions measured concentration level (e.g., "I could maintain a high concentration level while watching the video"), satisfaction with PC (e.g., "I was fully satisfied with the video I watched"), and self-confidence in the learning test (e.g., "I am very confident about my test results").

Results

We compared participant responses to the preliminary questionnaire variables and found no significant differences, confirming successful participant randomization. In particular, there were no significant differences between groups in media multitasking frequency, $F(1, 45 = .836$, $p = .366$; self-rated concentration ability, $F(1, 45) = .671$, $p = .47$; cognitive ability, $F(1, 45) = 1.582$, $p = .121$; or technology acceptance, $F(1, 45) =$

3.383, p = .073.

Effects on visual attention

Table 2 shows the proportion of visual attention to the primary and secondary screens and average switching frequency between the two for each cognitive load condition in the dual-screen group. SC took approximately 36% of visual attention from the high-cognitive-load (i.e., lecture) video and resulted in switching 24 times. SC also took 37% of visual attention from the low-cognitive-load (i.e., documentary) video with relatively low switching frequency (f = 14). However, for the medium-cognitive-load (i.e., conference presentation) video, visual attention was distributed evenly between screens (54% for the primary screen) with medium switching frequency (f = 20). There was a statistically nonsignificant effect of cognitive load level on the amount of visual attention on each screen, $F(2, 24) = 2.02$, $p = .155$. However, cognitive load level significantly influenced visual attention switching frequency, $F(2, 24) = 5.60$, $p = .010$, $\eta^2 = .564$.

[Insert Table 2 here.]

Effects on learning performance

A repeated-measures ANOVA with cognitive load level (low, medium, and high) and content relevance (low, medium, and high) as within-subjects factors and screen group (single and dual screen) as a between-subjects factor was performed to analyze learning performance. Figure 3 and Table 3 show each screen group's learning performance per

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cognitive load and relevance level. The dual-screen group's learning performance was inferior to the single-screen group's, but this difference was nonsignificant, $F(1, 44) = 2.79$, $p = .102$ (Figure 3[a]). Therefore, there was no evidence indicating that second-screen-application use affected PC learning.

[Insert Figure 3 and Table 3 here.]

Learning performance differed significantly per cognitive load level, $F(2, 88) = 7.96$, $p = .002$, $\eta^2 = .153$ (Figure 3[b]). The effect of the interaction between cognitive load level and screen group on learning performance was significant, $F(2, 88) = 4.08$, $p = .028$, $\eta^2 = .085$. In other words, the between-group difference in learning performance varied per cognitive load level: the single-screen group's performance was significantly superior to the dual-screen group's for the high-cognitive-load video, $t(44) = 2.71$, $p = .017$, $\eta^2 = .123$, but not the low-, $t(44) = .583$, $p = .563$, $\eta^2 = .008$, or medium-cognitive-load videos, $t(44) = -1.408$, $p = .300$, $\eta^2 = .024$.

Figure 3(c) shows that the main effect of SC relevance on learning performance was significant, $F(2, 88) = 12.56$, $p < .000$, $\eta^2 = .222$, but the interaction between SC relevance and screen group was nonsignificant, $F(2, 88) = 1.427$, $p = .246$, $\eta^2 = .031$. Specifically, the dual-screen group's learning performance was significantly inferior to the single-screen group's when SC relevance was low, $t(44) = 2.339$, $p = .035$, $\eta^2 = .097$, but not when it was medium, $t(44) = 1.091$, $p = .281$, $\eta^2 = .026$, or high, $t(44) = .461$, $p = .647$, $\eta^2 = .005$. In the planned comparison, we examined the effects of relevance on learning performance for each cognitive load condition (Table 3). For the dual screen group with high-cognitive-load video, medium, $F(1, 44) = 5.11$, $p = .029$, $\eta^2 = .104$, and low, $F(1, 44) = 5.30$, $p = .026$, $\eta^2 = .108$, SC relevance exerted a significantly negative effect on learning performance, compared to the single screen group. However, the dual-screen group's learning performance for the high-cognitive-load video was not significantly inferior to the single-screen group's when SC relevance was high, $F(1, 44) = 0.77$, $p = .386$, $\eta^2 = .017$. In addition, SC relevance did not exert a significant effect on learning performance for the low-cognitive-load video. Finally, unlike the high- and low-cognitive-load videos, the dual-screen group's learning performance for the medium-cognitive-load video was significantly superior to the single-screen group's when SC relevance was high, $F(1, 44) = 4.65$, $p = .037$, $\eta^2 = .004$.

Effects on user experience

An ANOVA with screen group as a between-subjects factor examined user experience. Second-screen application use exerted no significant effect on overall user experience between the single- ($M = 3.3$, $SD = 0.56$) and dual-screen ($M = 3.4$, $SD = 0.40$) groups, $F(1, 44) = .65$, $p = .42$, $\eta^2 = .015$. However, user experience differed significantly between groups per cognitive load level and question type (Table 4). For the high-cognitive-load video, self-confidence concerning the test in the dual-screen group ($M = 2.88$, $SD = 0.99$) was significantly lower, $F(1, 45) = 4.52$, $p = .04$, $\eta^2 = .092$, than the single-screen group's ($M = 3.50$, $SD = 0.95$). In addition, none of the responses concerning the low-cognitive-load video differed significantly between groups. For the medium-cognitive-load video, concentration levels and satisfaction with PC in the dual-screen group (M = 3.69, SD = 0.88 and M = 4.38, SD = 0.57, respectively) were significantly higher—$F(1, 45) = 6.45$, $p = .015$, $\eta^2 = .128$ and $F(1, 45) = 7.31$, $p = .010$, $\eta^2 = .143$, respectively—than those of the single-screen group (M = 2.95, SD = 1.10 and M = 3.65, SD = 1.23, respectively), consistent with learning performance. In summary, second-screen-application use exerted positive effects on participants' self-reported satisfaction with PC and concentration levels for the medium-cognitive-load video and a negative effect on self-confidence concerning the test for the high-cognitive-load video.

[Insert Table 4 here.]

Discussion

The study examined the comprehensive effects of second-screen-application use on attention, learning, and user experience in an educational context. Three hypotheses related to attention and learning and one related to user experience were tested. The results did not support H1, which suggested that second-screen use would decrease learning; however, H2, which suggested that the amount of cognitive load that PC requires would affect learning, was supported, implying that second-screen-application use does not degrade learning for some cognitive load conditions (i.e., low and medium). H3, which suggested that content relevance would differently affect PC learning, was not supported, but the dual-screen group's learning performance was significantly superior to the single-screen group's in certain conditions (e.g., medium-cognitive-load video with highly relevant SC). H4, which suggested that second-screen-application use would positively affect overall user experience, was partially supported, with the results under medium-cognitive-load condition. This partial support for the hypotheses has several implications for educational content in the practical design of second-screen applications.

Implications regarding attention and learning

Amount of cognitive load for PC was the most crucial factor affecting attention and learning in educational second-screen contexts. A significantly inferior learning performance was observed in the dual-screen group relative to the single-screen group's for the high-cognitive-load video with high visual attention switching frequency, even though attention to PC was maintained for longer than for SC. The results showed that the high-cognitive-load video led to visual attention high switching frequency, intended to capture PC and SC simultaneously. However, such efforts to retrieve information from two different screens (i.e., visual attention high switching frequency) inevitably resulted in cognitive overload, which exerted significantly detrimental effects on PC learning.

For the low-cognitive-load video, the non-significance of the effect of second-screen-application use on learning performance in the dual-screen group was attributed to low switching frequency and longer attention to PC. This result could be due to participants retrieving PC and SC simultaneously without overloading cognitive resources and following slow and descriptive plot structure. Consequently, users tended to focus on the video and follow the overall storyline rather than specific information provided on the secondary screen. This allowed users to switch visual attention to less informative scenes (e.g., scene transition) with low switching frequency without reducing learning performance.

In contrast, for the medium-cognitive-load video, the dual-screen group's learning performance was superior to the single-screen group's in certain conditions, although visual attention switching frequency was relatively high and the proportion of visual attention to the video was low. This occurred because the medium-cognitive-load video contained mainly audio (e.g., a presenter's speech) rather than visual information. As users tended to focus on auditory information in the medium-cognitive-load video, participants could follow the entire storyline using only the auditory information in PC while focusing visual attention on SC. This allowed participants to effectively allocate cognitive resources to different modalities and follow the modality principle (Moreno & Mayer, 1999) of multimedia learning, which states that a combination of pictures and audio narration better improves learning performance than a picture and on-screen text.

SC relevance also mediated the degree of integration between visual information retrieved from SC and continuously perceived auditory information from PC in working memory as locally influential factors. The dual-screen group's learning performance for the lecture and documentary videos was like the single-screen group's when SC relevance was high. This implies that SC relevance mediated learning, even in the high-cognitive-load condition. Furthermore, results concerning the conference presentation video showed that the dual-screen group's learning performance was superior to the single-screen group's when PC contained a larger amount of auditory than visual information, and SC content relevance was high. Therefore, SC relevance should be considered carefully, and SC should be designed for fine modulation of learning performance for each piece of PC.

Implications regarding user experience

User experience exhibited similar tendency with learning performance. For the high- and low-cognitive-load videos, the differences in concentration levels and satisfaction with PC between the dual- and single-screen groups were nonsignificant. However, the dual-screen group was less confident than the single-screen group regarding learning outcomes (i.e., the test), particularly for the high-cognitive-load video. This may be because they lacked confidence regarding retrieved and learned information as frequent visual attention switching caused them to miss certain parts of the visual information in the high-cognitive-load video, increasing uncertainty regarding the overall amount of information learned, regardless of actual retrieval.

However, the dual-screen group's concentration and satisfaction levels were higher than the single-screen group's for the medium-cognitive-load video. This result coincided with learning performance findings. Therefore, the above-mentioned characteristics of conference presentations were verified by the results regarding user experience.

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Design guidelines

Based on the above implications, second-screen-application use might not degrade learning or experience of PC and might even improve them when appropriate PC and SC are selected, combined, and designed carefully as video producers could affect attention and learning for videos via feature design (Anderson, Lorch, Field, Collins, & Nathan, 1986; Wright & Huston, 1983). We offer several design guidelines for educational second-screen applications based on our findings regarding general and specific perspectives. The following guidelines pertain to multiple content for second-screen applications:

- Carefully identify the typical plot structure, scene composition, and amount of visual and auditory information in PC, and determine the appropriateness of the chosen PC before applying it to second-screen applications.
- After choosing PC, design SC carefully, considering content relevance to maximize integration between auditory PC and SC rather than minimizing visual attention shift resulting from SC.
- Assign greater weight to auditory than to visual information in PC to maximize learning when SC is provided simultaneously.
- Second-screen application use is recommended, except when PC requires learning outcomes, to enhance user experience.

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The following specific design guidelines pertain to composition of multiple content for lecture, documentary, and conference presentation videos:

- For high-cognitive-load videos intended to deliver dense information, requiring high cognitive load, and aiming to motivate and encourage students, second-screen-application use should be avoided.
- Provision of SC is highly recommended for low-cognitive-load videos as it does not degrade learning or user experience.
- The provision of highly relevant SC is recommended for certain videos, which contain a larger amount of auditory than visual information, to enhance learning and user experience.

Contributions

The present study contributes to second-screen research in two ways. First, it demonstrated comprehensive behavioral, cognitive, and affective effects of second-screen-application use. Previous research focused on limited approaches, such as visual attention distribution (Breidbach et al., 2015; Holmes et al., 2012; Kusumoto et al., 2014) and user experience (Basapur et al., 2011; Kusumoto et al., 2014; Vinayagamoorthy et al., 2012); the current study examined the effects of second-screen-application use on attention, learning, and user experience in a single usage stream.

Second, the concept of content relevance, whereby practitioners could identify contextual relevance of SC to PC, was proposed. SC relevance to PC is crucial in

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second-screen applications intended to integrate multiple pieces of content into "synergized simultaneous streams of content" (Schultz, Block, & Raman, 2011). Previous research proposed message (Srivastava, 2013) and task relevance (Van Cauwenberge et al., 2014; Wang, Irwin, Cooper, & Srivastava, 2015), which are similar to content relevance in the current study. Message relevance concerns the personal relevance of the message to a user, and task relevance refers to the goal relevance between multiple tasks that a user performs. On the other hand, content relevance refers to contextual relevance of one content type to another. Message relevance is more appropriate to investigating effects of personal preferences. Task relevance is more appropriate for identifying effects of conventional media multitasking, in which it is assumed that multiple tasks are not necessarily related or relevant to each other, because the types of multiple tasks depend entirely on the user's intention whereas content relevance can be designed and selected by content providers. The content relevance concept could be used to scale contextual relevance levels, assuming all SC is already relevant to PC. Fine adjustment of content relevance levels could allow practitioners to improve second-screen-application design and evaluation.

Limitations and future research

Certain limitations should be noted in applying these guidelines to general second-screen applications. First, as we focused on passive processing of visual and auditory information in the second-screen application, to reduce excessive parameters, the effect of content interactivity was not considered. Second, we concentrated on educational

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rather than entertainment content, such as movies, drama, and reality shows, because of difficulty measuring learning performance in the latter. Third, learning pertained to PC rather than SC as we aimed to compare second- and single-screen-application effects. Furthermore, we focused on working memory rather than long-term learning as "the central work of multimedia learning takes place in working memory" (Mayer, 2005). Fourth, the sample size was relatively small and the participant gender proportion was skewed to male. However, participants were allocated equally for each screen group to balance the gender proportion. These limitations provide topics for further research. Future research should explore specific strategies via which second-screen applications, considering design issues such as interactivity and engagement, could better facilitate long-term learning than in single-screen environments.

Conclusion

This study examined the effects of second-screen-application use on attention, learning performance, and user experience per PC cognitive load and content relevance. The results showed that, when SC relevance was low for the high-cognitive-load video, learning performance was significantly lower than observed in a single-screen context. Interestingly, for the medium-cognitive-load video, which contains more information in the auditory than in the visual domain, learning performance increased significantly with highly relevant SC or strong integration of different types of visual and auditory information, even with little visual attention to PC. Moreover, second-screen-application use exerted little influence on user experience compared to learning performance. Based on these findings, we proposed design guidelines for composition of PC and SC to enhance learning performance and avoid distraction.

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No potential conflict of interest was reported by the authors.

References

Ainasoja, M., Linna, J., Heikkilä, P., Lammi, H., & Oksman, V. (2014). A Case Study on Understanding 2nd Screen Usage during a Live Broadcast-A Qualitative Multi-Method Approach. *The Eighth International Conference on Mobile Ubiquitous Computing, Systems, Services and Technologies.*

Albaili, M. A. (1997). Differences among low-, average-and high-achieving college students on learning and study strategies. *Educational Psychology*, 17(1-2), 171-177.

Anderson, D. R., Lorch, E. P., Field, D. E., Collins, P. A., & Nathan, J. G. (1986). Television viewing at home: Age trends in visual attention and time with TV. Child Development, 1024-1033.

Anderson, J. A. (1983). Television literacy and the critical viewer. In J. Bryant and D. R. Anderson, (Eds.) (1983). Children’s Understanding of Television: Research on attention and comprehension. New York: Academic Press.

Anstead, E., Benford, S., & Houghton, R. J. (2014). Many-screen viewing: evaluating an olympics companion application. *In Proc. of the 2014 ACM international* *conference on Interactive experiences for TV and online video* (pp. 103-110). ACM.

Armstrong, G. B., & Chung, L. (2000). Background Television and Reading Memory in Context: Assessing TV Interference and Facilitative Context Effects on Encoding Versus Retrieval Processes. *Communication Research*, 27(3), 327-352.

Baddeley, A. D., Scott, D., Drynan, R., & Smith, J. C. (1969). Short-term memory and the limited capacity hypothesis. *British Journal of Psychology*, 60(1), 51–55.

Basapur, S., Harboe, G., Mandalia, H., Novak, A., Vuong, V., & Metcalf, C. (2011). Field trial of a dual device user experience for iTV. *In Proc. of the 9th international interactive conference on Interactive television* (pp. 127-136). ACM.

Berk, R. A. (2009). Multimedia teaching with video clips: TV, movies, YouTube, and mtvU in the college classroom. *International Journal of Technology in Teaching and Learning*, 5(1), 1-21.

Bos, W. (2011). Javascript face detection + canvas + video = html5 glasses!, Webpage, http://wesbos.com/html5-video-face-detection-canvas-javascript.

Brasel, S. A., & Gips, J. (2011). Media Multitasking Behavior: Concurrent Television and Computer Usage. *Cyberpsychology, Behavior, and Social Networking*, 14(9), 527-534.

Breidbach, C. F., Chandler, J. D., & Maglio, P. P. (2015). The Duality of Second Screens: A Phenomenological Study of Multi-platform Engagement and Service Experiences. *In System Sciences (HICSS), 2015 48th Hawaii International Conference on* (pp. 1432-1441). IEEE.

Brown, A., Evans, M., Jay, C., Glancy, M., Jones, R., & Harper, S. (2014). HCI over multiple screens. *In CHI'14 Extended abstracts on human factors in computing systems* (pp. 665-674). ACM.

Carrier, L. M., Cheever, N. A., Rosen, L. D., Benitez, S., & Chang, J. (2009). Multitasking across generations: Multitasking choices and difficulty ratings in three generations of Americans. *Computers in Human Behavior*, 25(2), 483-489.

Chronister, P. (2014). *The Second Screen Landscape: Exploring the Motivations of Using Second Screen Devices While Watching TV* (Master's thesis). American University, Washington, DC.

Collins, R. L. (2008). Media multitasking: Issues posed in measuring the effects of television sexual content exposure. *Communication methods and measures*, 2(1-2), 65-79.

Courtois, C., & D'heer, E. (2012). Second screen applications and tablet users: constellation, awareness, experience, and interest. *In Proc. of the 10th European conference on Interactive tv and video* (pp. 153-156). ACM.

Cruickshank, L., Tsekleves, E., Whitham, R., Hill, A., & Kondo, K. (2007). Making Interactive TV Easier to Use: Interface Design for a Second Screen Approach. *The Design Journal*, 10(3), 41-53.

De Meulenaere, J., Staelens, N., Van den Broeck, W., De Turck, F., & Demeester, P. (2014). Evaluating the influence of second screen usage on video quality perception. *In Proc. of the 8th International workshop on Video Processing and Quality Consumer Electronics* (VPQM 2014).

Dell'Acqua, R., and Jolicoeur, P. (2000). Visual encoding of patterns is subject to dual-task interference. *Memory and Cognition*, 28(2), 184-191.

Entwistle, N.J. (1998). Understanding academic performance at university: A research retrospective. In D. Shorrocks (Ed.), Directions in educational psychology (pp. 106 – 127). London: Whurr.

Fante, R., Jacobi, L. L., & Sexton, V. D. (2013). The Effects of Instant Messaging and Task Difficulty on Reading Comprehension. *North American Journal of Psychology*, 15(2), 287.

Gamble, K. R., Howard Jr, J. H., & Howard, D. V. (2014). Does a simultaneous memory load affect older and younger adults' implicit associative learning?. *Aging, Neuropsychology, and Cognition*, 21(1), 52-67.

Geerts, D., Leenheer, R., De Grooff, D., Negenman, J., & Heijstraten, S. (2014). In front of and behind the second screen: viewer and producer perspectives on a companion app. *In Proc. of the 2014 ACM international conference on Interactive experiences for TV and online video* (pp. 95-102). ACM.

Hembrooke, H., & Gay, G. (2003). The laptop and the lecture: The effects of multitasking in learning environments. *Journal of Computing in Higher Education*, 15(1), 46-64.

Holmes, M. E., Josephson, S., & Carney, R. E. (2012). Visual attention to television programs with a second-screen application. *In Proc. of the symposium on eye tracking research and applications* (pp. 397-400). ACM.

Homer, B. D., Plass, J. L., & Blake, L. (2008). The effects of video on cognitive load and social presence in multimedia-learning. *Computers in Human Behavior*, 24(3), 786-797.

Jensen, J. A., Walsh, P., Cobbs, J., & Turner, B. A. (2015). The effects of second screen use on sponsor brand awareness: A dual coding theory perspective. *Journal of Consumer Marketing*, 32(2), 71-84.

Jeong, S. H., & Hwang, Y. (2015). Multitasking and persuasion: the role of structural interference. *Media Psychology*, 18(4), 451-474.

Jeong, S. H., & Hwang, Y. (2016). Media Multitasking Effects on Cognitive vs. Attitudinal Outcomes: A Meta-Analysis. *Human Communication Research*.

Jeong, S. H., & Hwang, Y. (2012). Does Multitasking Increase or Decrease Persuasion? Effects of Multitasking on Comprehension and Counterarguing. *Journal of Communication*, 62(4), 571-587.

Jolicoeur, P. (1999). Dual-task interference and visual encoding. *Journal of Experimental Psychology: Human Perception and Performance*, 25(3), 596-616.

Jolicoeur, P., and Dell'Acqua, R. (1999). Attentional and structural constraints on visual encoding., 62(2-3), 154-164.

Kahneman, D. (1973). Attention and effort (p. 246). Englewood Cliffs, NJ: Prentice-Hall.

Komarraju, M., & Nadler, D. (2013). Self-efficacy and academic achievement: Why do implicit beliefs, goals, and effort regulation matter?. *Learning and Individual Differences*, 25, 67-72.

Kusumoto, K., Kinnunen, T., Kätsyri, J., Lindroos, H., & Oittinen, P. (2014, November). Media experience of complementary information and tweets on a second screen. *In Proc. of the ACM International Conference on Multimedia* (pp. 437-446). ACM.

Laine-Hernandez, M., Kinnunen, T., Kusumoto, K., Kätsyri, J., Westman, S., Ravaja, N., & Oittinen, P. (2014). Multitasking influence on engagement. *Personal Media Day, Deliverable 2.3, 3*, 1-35.

Lang, A. (2000). The limited capacity model of mediated message processing. *Journal of communication*, 50(1), 46-70.

Lang, A. (2006). Using the limited capacity model of motivated mediated message processing to design effective cancer communication messages. *Journal of Communication*, 56, S57–S80.

Lang, A., Bolls, P., Potter, R. F., & Kawahara, K. (1999). The effects of production pacing and arousing content on the information processing of television messages. *Journal of Broadcasting & Electronic Media*, 43(4), 451-475.

Lang, A., Kurita, S., Gao, Y., & Rubenking, B. (2013). Measuring television message complexity as available processing resources: Dimensions of information and cognitive load. *Media Psychology*, 16(2), 129-153.

Lang, A., Park, B., Sanders-Jackson, A. N., Wilson, B. D., & Wang, Z. (2007). Cognition and emotion in TV message processing: How valence, arousing content, structural complexity, and information density affect the availability of cognitive resources. *Media Psychology*, 10(3), 317-338.

Lee, J., Lin, L., & Robertson, T. (2011). The impact of media multitasking on learning. *Learning, Media and Technology*, 37(1), 94-104.

Lee, S., Yoo, J., & Han, G. (2015, January). A second screen web service framework based on multimedia package distribution. *In Consumer Electronics (ICCE), 2015 IEEE International Conference on* (pp. 42-45). IEEE.

Liaw, S. S., & Huang, H. M. (2013). Perceived satisfaction, perceived usefulness and interactive learning environments as predictors to self-regulation in e-learning environments. *Computers & Education*, 60(1), 14-24.

Lin, L., Robertson, T., & Lee, J. (2009). Reading Performances between Novices and Experts in Different Media Multitasking Environments. *Computers in the Schools*, 26(3), 169-186.

Loranger, A. L. (1994). The study strategies of successful and unsuccessful high school students. *Journal of Literacy Research*, 26(4), 347-360.

Maki, W. S., & Maki, R. H. (2002). Multimedia comprehension skill predicts differential outcomes of web-based and lecture courses. *Journal of Experimental Psychology: Applied*, 8(2), 85.

Mayer, R. E. (2005). *Principles for reducing extraneous processing in multimedia learning: Coherence, signaling, redundancy, spatial contiguity, and temporal contiguity principles*. The Cambridge handbook of multimedia learning, 183-200.

Mayer, R. E. (2009). *Multimedia learning*. Cambridge university press.

Mayer, R. E., & Moreno, R. (1998). A split-attention effect in multimedia learning: Evidence for dual processing systems in working memory. *Journal of Educational Psychology*, 90(2), 312-320.

Mayer, R. E., & Moreno, R. (2003). Nine ways to reduce cognitive load in multimedia learning. *Educational psychologist*, 38(1), 43-52.

Moreno, R., & Mayer, R. E. (1999). Cognitive principles of multimedia learning: The role of modality and contiguity. *Journal of Educational Psychology*, 91(2), 358-368.

Morony, S., Kleitman, S., Lee, Y. P., & Stankov, L. (2013). Predicting achievement: Confidence vs self-efficacy, anxiety, and self-concept in Confucian and European countries. *International Journal of Educational Research*, 58, 79-96.

Murdock Jr, B. B. (1962). The serial position effect of free recall. *Journal of experimental psychology*, 64(5), 482.

Neate, T., Jones, M., & Evans, M. (2015, April). Mediating attention for second screen companion content. *In Proc. of the SIGCHI conference on human factors in computing systems* (pp. 3103-3106). ACM.

Nielsen Company, Action figures: How second screens are transforming tv viewing. Nielsen Report, http://www.nielsen.com/us/en/insights/news/2013/action-figures--how-second-screens-are-transforming-tv-viewing.html (2013).

Oviedo, V., Tornquist, M., Cameron, T., & Chiappe, D. (2015). Effects of media multi-tasking with Facebook on the enjoyment and encoding of TV episodes. *Computers in Human Behavior*, 51, 407-417.

Park, B., Moreno, R., Seufert, T., & Brünken, R. (2011). Does cognitive load moderate the seductive details effect? A multimedia study. *Computers in Human Behavior*, 27(1), 5-10.

Rideout, V. J., Foehr, U. G., & Roberts, D. F. (2010). *Generation M2: Media in the Lives of 8- to 18-Year-Olds.* Menlo Park, CA: Henry J. Kaiser Family Foundation.

Roca, J. C., Chiu, C.-M., & Martínez, F. J. (2006). Understanding e-learning continuance intention: An extension of the Technology Acceptance Model. *International Journal of human-computer studies*, 64(8), 683-696.

Salvucci, D. D., & Bogunovich, P. (2010, April). Multitasking and monotasking: the effects of mental workload on deferred task interruptions. *In Proc. of the SIGCHI conference on human factors in computing systems* (pp. 85-88). ACM.

Salvucci, D. D., & Taatgen, N. A. (2008). Threaded cognition: an integrated theory of concurrent multitasking. *Psychological review*, 115(1), 101.

Sana, F., Weston, T., & Cepeda, N. J. (2013). Laptop multitasking hinders classroom learning for both users and nearby peers. *Computers & Education*, 62, 24-31.

Schultz, D. E., Block, M. P., & Raman, K. (2011). Understanding consumer-created media synergy. *Journal of Marketing Communications*, 18(3), 173-187.

Shah, P., & Freedman, E. G. (2003). Visuospatial cognition in electronic learning. *Journal of Educational Computing Research*, 29(3), 315-324.

Shim, H., Oh, P., Song, H., & Lee, Y. (2015). An exploration of motivations for two screen viewing, social interaction behaviors, and factors that influence viewing intentions. *Cyberpsychology, Behavior, and Social Networking*, 18(3), 158-164.

Shin, D. H., An, H., & Kim, J. H. (2015). How the second screens change the way people interact and learn: the effects of second screen use on information processing. *Interactive Learning Environments*, 1-22.

Srivastava, J. (2013). Media multitasking performance: Role of message relevance and formatting cues in online environments. *Computers in Human Behavior*, 29(3), 888-895.

Stankov, L., & Lee, J. (2014). Quest for the best non-cognitive predictor of academic achievement. *Educational Psychology*, 34(1), 1-8.

Stankov, L., Lee, J., Luo, W., & Hogan, D. J. (2012). Confidence: A better predictor of academic achievement than self-efficacy, self-concept and anxiety? *Learning and Individual Differences*, 22(6), 747-758.

Steinberg, B., Study: Young Consumers Switch Media 27 Times an Hour., Advertising Age, http://adage.com/article/news/study-young-consumers-switch-media-27-times-hour/234008/ (2012).

Stroop, J. R. (1935). Studies of interference in serial verbal reactions. Journal of experimental psychology, 18(6), 643.

Taneja, S., & Goel, A. (2014). MOOC providers and their strategies. *International Journal of Computer Science and Mobile Computing,* 3 (5), 222-228.

The 2nd Screen Society, Lexicon for the 2nd screen society, Webpage, http://www.2ndscreensociety.com/lexicon/#SecondScreen (2012).

Tran, P., Carrillo, R., & Subrahmanyam, K. (2013). Effects of online multitasking on reading comprehension of expository text. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace*, 7(3).

Tulving, E., Mandler, G., & Baumal, R. (1964). Interaction of two sources of information in tachistoscopic word recognition. Canadian Journal of Psychology/Revue canadienne de psychologie, 18(1), 62.

Van Cauwenberge, A., Schaap, G., & van Roy, R. (2014). "TV no longer commands our full attention": Effects of second-screen viewing and task relevance on cognitive load and learning from news. *Computers in Human Behavior, 38(0),* 100-109.

Vinayagamoorthy, V., Allen, P., Hammond, M., & Evans, M. (2012, May). Researching the user experience for connected tv: a case study. *In CHI'12 Extended Abstracts on Human Factors in Computing Systems* (pp. 589-604). ACM.

Voorveld, H. A. M. (2011). Media multitasking and the effectiveness of combining online and radio advertising. *Computers in Human Behavior*, 27(6), 2200-2206.

Wang, Z., David, P., Srivastava, J., Powers, S., Brady, C., D'Angelo, J., & Moreland, J. (2012). Behavioral performance and visual attention in communication multitasking: A comparison between instant messaging and online voice chat. *Computers in Human Behavior*, 28(3), 968-975.

Wang, Z., Irwin, M., Cooper, C., & Srivastava, J. (2015). Multidimensions of media multitasking and adaptive media selection. *Human Communication Research*, *41*(1), 102-127.

Wright, J. C., & Huston, A. C. (1983). A matter of form: Potentials of television for young viewers. American Psychologist, 38(7), 835.

Yip, M. C., & Chung, O. L. (2005). Relationship of study strategies and academic performance in different learning phases of higher education in Hong Kong. *Educational research and evaluation*, 11(1), 61-70.

Zhang, W., Jeong, S.-H., & Fishbein, M. (2010). Situational factors competing for attention: The interaction effect of multitasking and sexually explicit content on TV recognition. *Journal of Media Psychology: Theories, Methods, and Applications*, 22(1), 2-13.

Zhu, C. (2012). Student Satisfaction, Performance, and Knowledge Construction in Online Collaborative Learning. *Educational Technology & Society*, 15(1), 127-136.

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Tables

Table 1. Content relevance levels and rank according to cognitive load.

		Primary content		
		Low	Medium	High
Secondary content	Personal information	**M** (4)	**H** (1)	**H** (2)
	Related news	**H** (2)	**H** (3)	**M** (5)
	Episode list	**M** (6)	**L** (7)	**H** (3)
	Academic references	**H** (1)	**H** (2)	**H** (1)
	Background information	**H** (3)	**M** (5)	**L** (9)
	Advertisement	**L** (9)	**L** (9)	**L** (8)
	Price information	**L** (8)	**L** (8)	**M** (6)
	Common SNS reaction	**M** (5)	**M** (4)	**M** (4)
	Celebrities' SNS reaction	**L** (7)	**M** (6)	**L** (7)

H = high relevance; **M** = medium relevance; **L** = low relevance; ( ) = rated rank

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Table 2. Visual attention distribution and switching frequency in dual-screen group according to cognitive load.

Cognitive load	Screen	Visual attention (%)	Switching frequency
Low	Primary screen	63	14
	Secondary screen	37	
Medium	Primary screen	54	20
	Secondary screen	46	
High	Primary screen	64	24
	Secondary screen	36	

Table 3. Learning performance per content relevance for each cognitive load condition.

Cognitive load	Content relevance	Single			Dual			df	F	η^2	p
		M	SD	SE	M	SD	SE				
Low	High	0.72	0.23	0.05	0.62	0.23	0.04	1	2.05	.045	0.16
	Medium	0.73	0.19	0.04	0.77	0.18	0.04	1	0.65	.015	0.42
	Low	0.85	0.24	0.05	0.85	0.27	0.05	1	0.00	.000	0.96
Medium	**High**	**0.66**	**0.18**	**0.04**	**0.75**	**0.12**	**0.02**	**1**	**4.65**	**.096**	**0.04***
	Medium	0.92	0.09	0.02	0.88	0.13	0.03	1	1.50	.033	0.23
	Low	0.79	0.22	0.05	0.76	0.18	0.04	1	0.17	.004	0.69
High	High	1.00	0.00	0.00	0.96	0.20	0.04	1	0.77	.017	0.39
	Medium	**0.89**	**0.13**	**0.03**	**0.77**	**0.22**	**0.04**	**1**	**5.11**	**.104**	**0.03***
	Low	**0.93**	**0.13**	**0.03**	**0.75**	**0.32**	**0.06**	**1**	**5.30**	**.108**	**0.03***

*** p < .05**

M = mean; SD = standard deviation; SE = standard error

Table 4. User experience for each cognitive load condition.

		Single			Dual			df	F	η^2	p
		M	SD	SE	M	SD	SE				
Low	Concentration level	2.85	1.23	0.19	2.92	0.98	0.17	1	0.05	.001	0.82
	Satisfaction with PC	3.25	1.07	0.23	3.77	0.99	0.19	1	2.89	.062	0.09
	Self-confidence regarding test	2.60	1.10	0.24	2.19	0.75	0.14	1	2.24	.049	0.14
Medium	**Concentration level**	**2.95**	**1.10**	**0.27**	**3.69**	**0.88**	**0.19**	**1**	**6.45**	**.128**	**0.01***
	Satisfaction with PC	**3.65**	**1.23**	**0.27**	**4.38**	**0.57**	**0.11**	**1**	**7.31**	**.143**	**0.01***
	Self-confidence regarding test	3.05	1.10	0.24	2.81	0.75	0.14	1	0.79	.018	0.38
High	Concentration level	3.80	0.89	0.19	3.65	0.89	0.17	1	0.30	.007	0.59
	Satisfaction with PC	4.05	0.76	0.17	4.42	0.70	0.14	1	2.97	.063	0.09
	Self-confidence regarding test	**3.50**	**0.95**	**0.21**	**2.88**	**0.99**	**0.19**	**1**	**4.52**	**.093**	**0.04***

*** p < .05**

M = mean; SD = standard deviation; SE = standard error

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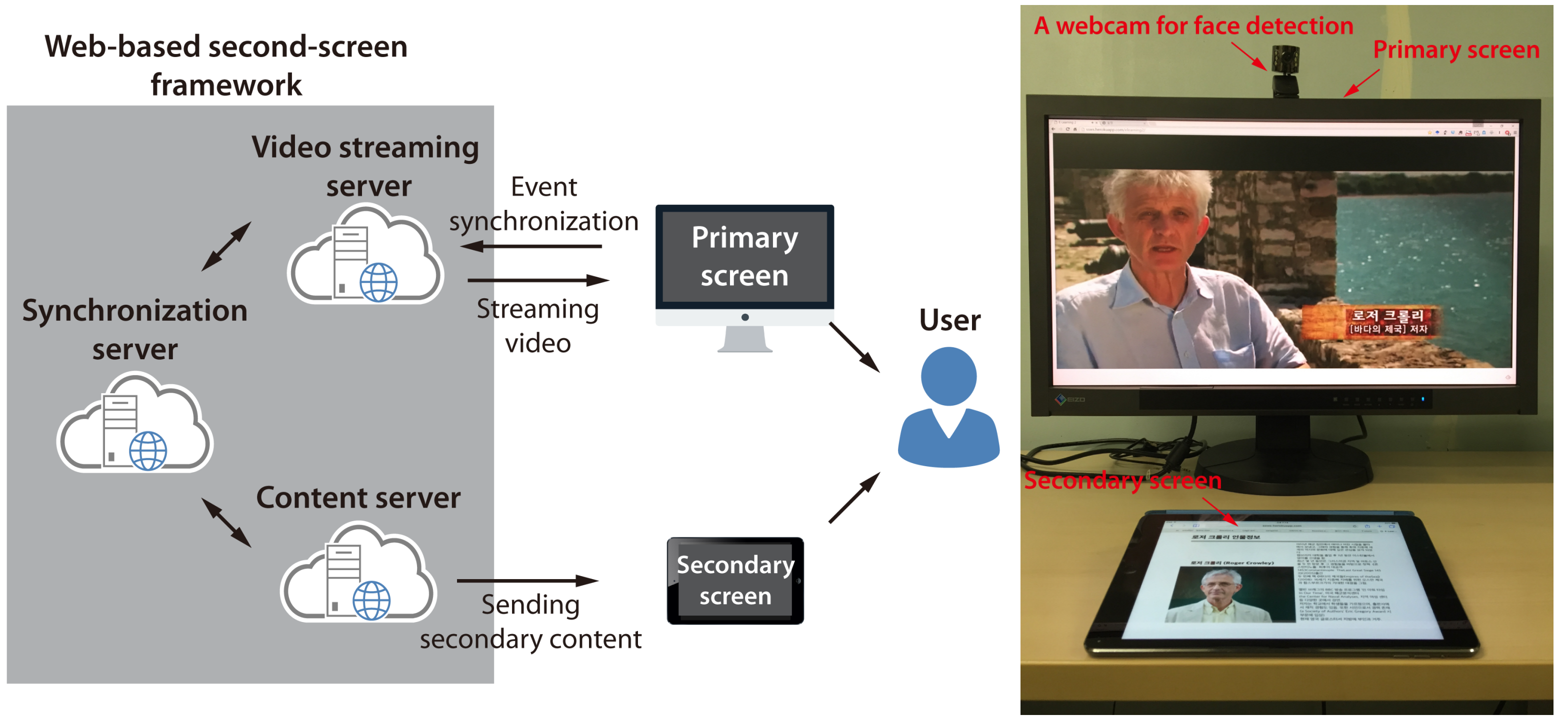
Figures



(a) (b)

Figure 1. Experimental environment; (a) web-based second-screen framework; (b) equipment.

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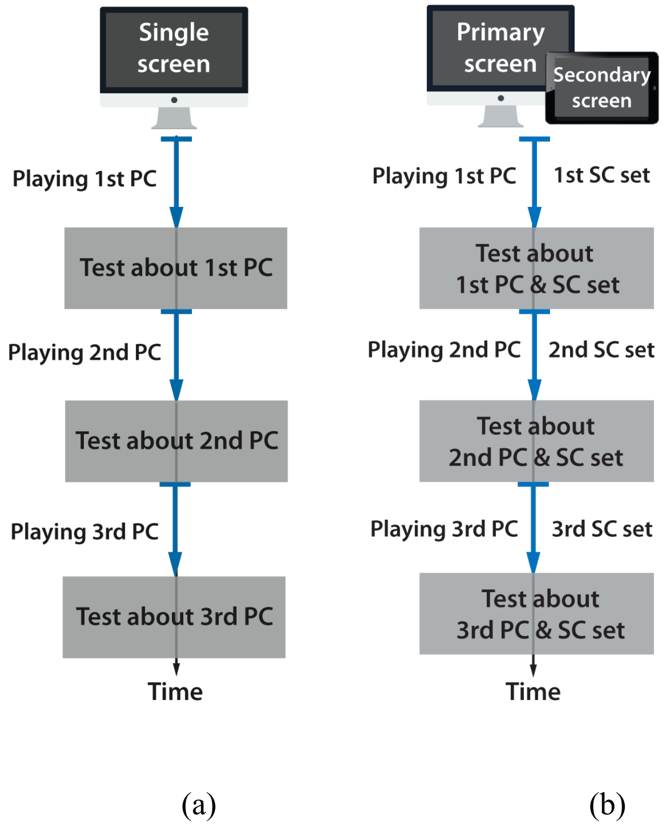

Figure 2. (a) Experimental procedure for the single-screen group; (b) experimental procedure for the dual-screen group. PC: primary content; SC: secondary content.

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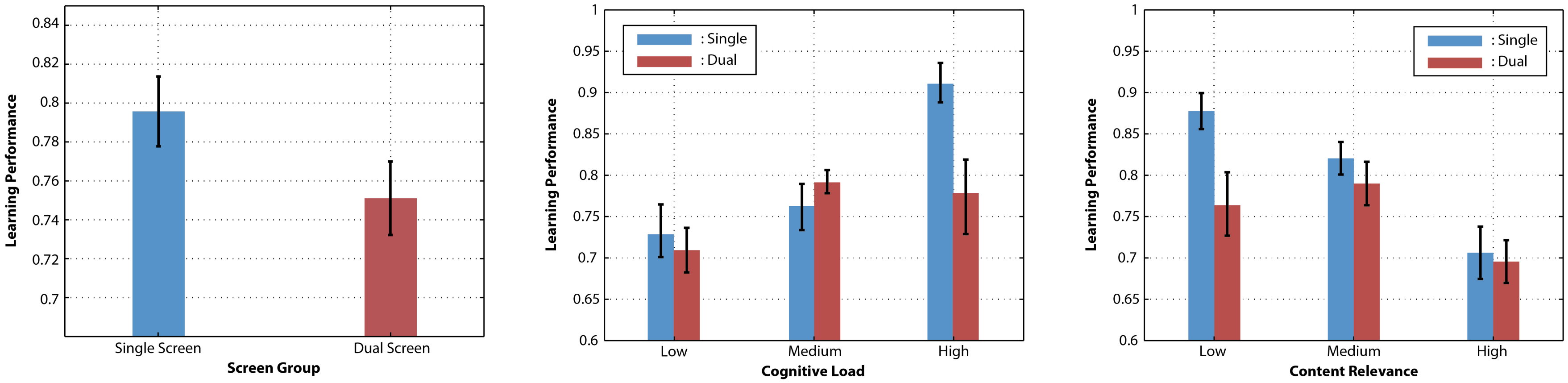

(a) (b) (c)

Figure 3. (a) Learning performance between screen groups; (b) learning performance per cognitive load level; (c) learning performance per content relevance.